United International University
Department of Computer Science and Engineering
CSE 4531: Computer Security
Final Examination: Fall 2025
Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. Numbers to the right of the questions denote their marks.

1(a)	A startup company creates a proprietary encryption system and refuses to disclose its design, claiming that secrecy makes it more secure. The system is also highly complex, requiring many interdependent modules. Which security principles are violated in this scenario? How do these approaches increase security risk rather than reducing them?	[5]
1(b)	How does botnets work? Explain with a figure. What types of attacks can be launched using botnets? List few countermeasures	[5]
2(a)	Perform encryption and decryption of the plaintext 9 using RSA algorithm for p = 5, q = 11 and e = 3.	[5]
2(b)	Explain the philosophy behind Diffie-Hellman key exchange protocol. In the Diffie-Hellman key exchange protocol, let the prime number be 353 and one of its prime root is 3. Let the users A and B select their secret keys 97 and 233, respectively. Compute the public keys of A and B and also their common secret key.	[5]
3(a)	What would be the financial advantage for a malware designer to create lots of different malicious code instances that all exploit the same vulnerability yet have different malware signatures?	[5]
3(b)	A university network experiences a sudden surge in traffic as hundreds of machines begin scanning and infecting other unpatched systems automatically. Using the worm propagation model, explain how the values of susceptible hosts (S(t)) and infected hosts (I(t)) change over time.	[5]
4(a)	What is the difference between ***SYN flood*** attack and ***Smurf*** attack? Explain the process to defend against SYN flooding attack using SYN cookie.	[5]
4(b)	Distinguish between ***Click-jacking*** and ***Cross Site Scripting(XSS)*** attacks. How can XSS lead to ***Session Hijacking***?	[5]